Table 4

Comparison of current network attack-type classifiers

Classifier	Overall ten-class classification				DoS		Exploits
	Accuracy	Macro F1	Balanced accuracy	Weighted F1	Recall	F1	Recall
XGBoost V1	0.8287	0.5979	0.5918	0.8045	0.0603	0.1113	**0.9497**
Weighted XGBoost V2	0.8168	**0.6431**	**0.6551**	**0.8234**	**0.6391**	**0.4645**	0.6646
Conflict-aware XGBoost	**0.8306**	0.6137	0.6036	0.8091	0.0823	0.1443	0.9422
CatBoost	0.8209	0.5607	0.5388	0.8012	0.1046	0.1704	0.9227

Notes: All metrics are fractions between 0 and 1. Bold values denote the largest observed value in each metric column.

UNSW-NB15 development validation (35,167 records, ten classes). Macro F1 and balanced accuracy help expose poor performance on minority attack classes.

Overall metrics come from saved model reports. Original XGBoost per-class scores were recalculated from its saved confusion matrix;
other per-class values are rounded diagnostics transcribed from earlier experiment output. These values were not independently recomputed here.

The development-validation fold was examined repeatedly; these comparisons do not represent an independent final TEST-set assessment.